15 Alison has some shapes.

She has 14 red cubes and 10 red spheres.
She has 12 black cubes and 8 black spheres.

Alison is going to select 2 of these shapes.

Of these 2 shapes

only 1 can be red
only 1 can be black
only 1 can be a cube
and only 1 can be a sphere.

In how many ways can Alison select the 2 shapes?

Red cube and black sphere: $14 \times 8 = 112$

1 mark
Multiply either valid pair.

Red sphere and black cube: $10 \times 12 = 120$

Total = $112 + 120 = 232$

1 mark
Add both mutually exclusive cases.

(Total for Question 15 is 2 marks)